

COMPUTER SCIENCES

CS 368-1 (2011 Summer) — Day 7 Homework

Due Friday, July 22, at the start of class.

Description

Read two input data files containing data about the countries of the world and store the data into useful data structures. Then, ask for user input and print a simple report based on the stored data.

Details

Your script should be divided into two main parts:

1. Reading and storing the country data
2. Reporting on that data

Phase I: Reading and Storing Data

There are two separate data files containing country data. The first file is called [input-07-country.txt](#), and looks like this:

```
ABW : Aruba : Aruba : Latin America & Caribbean
ADO : Andorra : Principality of Andorra : Europe & Central Asia
AFG : Afghanistan : Islamic State of Afghanistan : South Asia
AGO : Angola : People's Republic of Angola : Sub-Saharan Africa
ALB : Albania : Republic of Albania : Europe & Central Asia
```

That is, there are 4 data fields, separated by the string " : " (space - colon - space):

- Country code
- Short country name
- Long country name
- Geographic region

The second data file is called [input-07-population.txt](#), and looks like this:

```
ABW : 1960 : 49205
ABW : 1961 : 50244
ABW : 1962 : 51258
ABW : 1963 : 52224
ABW : 1964 : 53117
```

There are 3 data fields, separated by the string " : " (space - colon - space):

- Country code

- Year
- Population

Read both data files and store all of the data in well-considered data structures. Some hints:

- To parse each input line, use the `split` function:

```
my @line_elements = split(' : ', $input_line);
```

It is the opposite of the `join` function: It takes a single string, and splits the string into separate array elements by removing any instances of the first argument. It returns the array of elements found. Look it up on Perldoc, if you want. Yes, technically, the first argument is a regular expression, but the string argument given above works just fine for this assignment.

- This is real data, so it is a bit messy. There are “extra” countries in the population file. Only store the data for the countries that occur in both files.
- Read the descriptions of the reports below **first**. Think about how you will need to access the data to create your report. It may influence how you store the data.
- The population data is from 1960–2002, generally. However, the world changes, data are sometimes unavailable. Thus, the population statistic is **not** available for every country for every year. Be careful about your assumptions when building the data structures.

Phase II: Reports

Now that you have data, we can analyze it a bit and produce reports based on user input.

Note: Pick just **one** of the following reports! They are listed below in roughly increasing order of difficulty.

Report 1: World Population for a Year

Ask the user for a year, then calculate the total world population for that year. Sample output:

```
Calculate world population for what year? 1970
World population in 1970 was 3665297114 (202 countries).
```

```
Calculate world population for what year?
```

Report 2: Population Growth for a Country

Ask the user for a country **code**, start year, and end year. Calculate and display the population growth in that country from the start year to end year, both as an absolute amount and percentage growth; display the short or long name for the country, not the code. Sample output:

```
Calculate population growth for what country code? JPN
Starting in what year? 1970
Ending in what year? 2000
From 1970-2000, Japan grew by 22525000 (21.6%).
```

Calculate population growth for what country code?

Report 3: Top Population Growth

Ask the user for a start year and end year. Calculate and display the top N (5–10 is best) countries based on growth (either absolute growth or percentage growth, your pick). Display short or long names for countries, not codes. Sample output:

```
Starting in what year? 1970
Ending in what year? 2000
From 1970-2000, the following countries grew the most:
468354000 India
444330000 China
88359301 Indonesia
78183086 Brazil
77473000 Pakistan
77120000 United States
71588515 Bangladesh
68374892 Nigeria
47370000 Mexico
41122025 Philippines
```

Starting in what year?

Extra Challenges

- [Medium] Support more than one report. Present a menu of report options for the user, each time through the main loop. How do you organize your code effectively?
- [Hard] Search the Internet for a technique that displays large integers with commas: e.g., 123456789 → 123,456,789. Integrate the technique into your report(s).
- [Very Hard] Notice that “region” field of the first data file? Rewrite any above to calculate and display statistics for **regions** instead. For Report #1, report on just one region. For Report #1 or #2, there are no codes for regions (in our data file), so present the user with a list of regions to select from. Does this change in the reporting also change how you arrange your data structures?

Reminders

Do the work yourself, consulting reasonable reference materials as needed; any reference material that gives you a complete or nearly complete solution to this problem or a similar one is not OK to use. Asking the instructors for help is OK, asking other students for help is not.

Hand In

A **printout** of your code on a **single sheet of paper**. Be sure to put your own name in the initial comment block of the code. Identifying your work is important, or you may not receive appropriate credit.